# Hanping Xu

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## Personal Data


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| Citizenship and Year of Birth: | Chinese \| 1996 |
| email: | xuhanping@u.nus.edu |
| Homepage: | https://maimaidou.github.io/HanpingXu.github.io/ |


## Education

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| 2018 - 2023 | Ph.D. in Mathematical Economics, **National University of Singapore** (NUS), Singapore<br>Advisor: Prof. Yeneng Sun GPA: 4.89/5.0 |
| 2014 - 2018 | B.S. in Science with Honors Degree, **Sun Yat-sen University** (SYSU), Guangzhou, China<br>Major: Statistics |

## Working experience

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| 2025.1 - present | Postdoctoral Research Fellow, **National University of Singapore**(NUS), Singapore |
| 2024.3 - 2024.12 | Postdoctoral Research Fellow, **The Chinese University of Hong Kong**(CUHK), Hong Kong |
| 2023.1 - 2023.12 | Research Assistant, **The Chineses University of Hong Kong**(CUHK), Hong Kong, China |
| 2022.8 - 2022.12 | Student Research Assistant, **National University of Singapore**(NUS), Singapore |

## Research interests

Game Theory, Microeconomic Theory, Mathematical Economics

## Publication

**Equilibrium convergence in large games** (with Enxian Chen and Bin Wu), Journal of Mathematical Economics, Available online 7 February 2025, 103097.

**Pareto-undominated and socially-maximal Nash equilibria with coarser traits** (with Bin Wu), Economics Letters, 215(2022) : 110464.

## Working Papers

**Monotone Perfection**
with Wei He and Yeneng Sun

Abstract: Monotone equilibria may be undesirable in Bayesian games in the sense that players adopt weakly dominated strategies. To account for the possibility that the players might choose unintended strategies through a trembling hand, we propose an equilibrium refinement called "perfect monotone equilibrium." This notion strengthens the notion of monotone equilibrium in the sense that it satisfies the important property of admissibility in Bayesian games with finitely many actions, and the property of limit undominatedness in Bayesian games with infinitely many actions.

In a general class of Bayesian games where each player's action set is a sublattice of multi-dimensional Euclidean space and players' types are also multi-dimensional, a perfect monotone equilibrium is shown to exist under the supermodularity and increasing differences conditions. These conditions model the scenarios in which, informally, players' payoffs have complementarity in own actions and monotone incremental returns in own types. We demonstrate that the increasing differences condition is sharp by providing a two-player game that satisfies the widely adopted single crossing condition in the literature, but does not possess any perfect monotone equilibrium. To show the usefulness of our result in the setting with discontinuous payoffs, we provide various illustrative applications, including first-price auctions, all-pay auctions, and Bertrand competitions. Our result can be further extended to the setting with more general action spaces and type spaces.

**The equilibrium property of obvious strategy profiles in games with many players**

with Enxian Chen and Bin Wu

Abstract: This paper studies the equilibrium properties of the "obvious strategy profile" in large finite-player games. Each player in such a strategy profile simply adopts a randomized strategy as she would have used in a symmetric equilibrium of an idealized large game. We show that, under a continuity assumption, (i) obvious strategy profiles constitute a convergent sequence of approximate symmetric equilibria as the number of players tends to infinity, and (ii) realizations of such strategy profiles also form a convergent sequence of (pure strategy) approximate equilibria with probability approaching one. Our findings offer a solution that is easily implemented without coordination issues and is asymptotically optimal for players in large finite games. Additionally, we present a convergence result for approximate symmetric equilibria.

**Does Randomization Matter in Board Games?**

with Enxian Chen, Wei He, and Yeneng Sun

Abstract: This paper investigates whether players can gain an advantage by using randomized strategies in board games. To formalize the idea that behavioral and pure strategy subgame perfect equilibria (SPEs) lead to the same outcomes, we introduce the concept of no-mixing property: any realized strategy profile of a behavioral strategy SPE must also be a pure strategy SPE. Our first result establishes that the no-mixing property holds in two-player zero-sum sequential games, including many board games, even with the presence of Nature, general action spaces, and infinite horizons. We provide counterexamples demonstrating that this property fails without the "two-player" or the "zero-sum" assumptions. For general multi-player sequential games, we identify a sufficient condition – the single payoff condition – under which the no-mixing property holds. This condition reflects the observation that when one player makes a winning move in a board game, all other players lose. Finally, we present two applications in two-player zero-sum sequential games with Nature: a new existence result for pure strategy SPEs and the construction of a universal SPE that captures all pure strategy SPEs.

**Games with incomplete information: a general purification result**

with Wei He and Yeneng Sun

Abstract: We present a new purification result for Bayesian games with countably many actions, interdependent payoffs and correlated types. It is shown that the condition of coarser inter-player information characterizes the existence of purification, and also the existence of pure strategy equilibrium in these games. We demonstrate that the condition of countably many actions is tight for the purification result and pure strategy equilibrium existence. To prove the results for Bayesian games, we provide a general purification principle, which covers various earlier results as special cases.

**Large games with coarser traits and countable actions**

Abstract: We show that the coarser traits condition is both necessary and sufficient for the idealized limit property of large games with traits and countable actions. Meanwhile, we also show that the coarser traits condition is both necessary and sufficient for the existence of pure strategy Pareto-undominated socially-maximal Nash equilibria in large games with traits and countable actions. We demonstrate that the condition of countably many actions is tight.

## Teaching Experience

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| *National University of Singapore* | MA4264: Game Theory | 2021/2022 SEM 2 |
| | MA4264: Game Theory | 2020/2021 SEM 2 |
| | MA3238: Stochastic Processes I | 2019/2020 SEM 2 |
| | MA2216: Probability | 2019/2020 SEM 1 |
| *Teaching evaluation score* | 4.3/5.0 for Game Theory | 4.1/5.0 for Stochastic Process I |

## Presentations

2024 The Seventh World Congress of the Game Theory Society
Presenter for "Monotone Perfection"

2024 China Meeting on Game Theory and Its Applications
Presenter for "Obvious approximate symmetric equilibrium in games with many players"

2022 Society for the Advancement of Economic Theory (SAET) Conference
Presenter for "Obvious approximate symmetric equilibrium in games with many players"

2021 Society for the Advancement of Economic Theory (SAET) Conference
Presenter for "Games with incomplete information: a general purification result"

## Selected Scholarships, Honors and Awards

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| Part-time teaching assistant award, Faculty of Science, NUS | 2022 |
| Research Scholarships, NUS | 2019, 2020, 2021, 2022 |
| Top Graduate Tutor, Department of Mathematics, NUS | 2021 |
| Outstanding Graduate Award, Sun Yat-sen University | 2018 |
| National Scholarships, China | 2015, 2016, 2017 |
| The First Prize Scholarships, Sun Yat-sen University | 2015, 2016, 2017 |

## Reviewer for Journal

Economic Theory, Journal of Mathematical Economics

## References


*Professor Yeneng Sun*
Departments of Economics and Mathematics
National University of Singapore
Tel: +65 6516 3994
Email: ynsun@nus.edu.sg

*Associate Professor Wei He*
Departments of Economics
The Chinese University of Hong Kong
Tel: (852) 3943 8196
Email: hewei@cuhk.edu.hk

*Associate Professor Xiao Luo*
Department of Economics
National University of Singapore
Tel: +65 6516 6231
Email: ecslx@nus.edu.sg